

MICRO

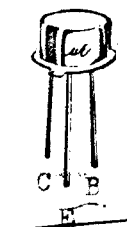
2N2193

NPN

SILICON
TRANSISTOR

2N2193 is NPN silicon planar transistor designed for medium power switching and amplifier applications.

T0-39



ABSOLUTE MAXIMUM RATINGS

Collector-Base Voltage	VCBO	80V
Collector-Emitter Voltage	VCEO	50V
Emitter-Base Voltage	VEBO	8V
Collector Current	IC	1A
Total Power Dissipation	Ptot	800mW
Operating Junction & Storage Temperature	Tj, Tstg	-65 to +200°C

ELECTRICAL CHARACTERISTICS (TA=25°C unless otherwise specified)

PARAMETER	SYMBOL	MIN	MAX	UNIT	TEST CONDITIONS
Collector-Base Breakdown Voltage	BVCBO	80		V	IC=100μA IE=0
Collector-Emitter Breakdown Voltage	LVCEO	50		V	IC=10mA IB=0
Emitter-Base Breakdown Voltage	BVEBO	8		V	IE=100μA IC=0
Collector Cutoff Current	ICBO		10	nA	VCB=60V IE=0
			25	μA	VCB=60V TA=150°C
Emitter Cutoff Current	IEBO		50	nA	VEB=5V IC=0
D.C. Current Gain	HFE	15			VCE=10V IC=100μA
		30			VCE=10V IC=10mA
		40	120		VCE=10V IC=150mA*
		20			VCE=10V IC=500mA*
		15			VCE=10V IC=1A*
		30			VCE=1V IC=150mA*

MICRO ELECTRONICS LTD. 美科有限公司

38 Hung To Road, Kwun Tong, Kowloon, Hong Kong. Cable: Microtron, Hong Kong. Telex: 43510 Micro Hx.
P.O. Box 69477, Kwun Tong. Tel: 3-430181-6, 3-893363, 3-892423, 3-898221 FAX: 3-410321

PARAMETER	SYMBOL	MIN	MAX	UNIT	TEST CONDITIONS
Base-Emitter Saturation Voltage	$V_{BE(sat)}$		1.3	V	$I_C=150mA$ $I_B=15mA^*$
Collector-Emitter Saturation Voltage	$V_{CE(sat)}$		0.35	V	$I_C=150mA$ $I_B=15mA^*$
Small Signal Current Gain	h_{fe}	2.5			$V_{CE}=10V$ $I_C=50mA$ $f=20MHz$
Output Capacitance	C_{ob}		20	pF	$V_{CB}=10V$ $f=1MHz$

* Pulse Test : Pulse Width = $300\mu S$, Duty Cycle $\leq 2\%$.